Indian Institute of Technology Bombay

PH 544 - Quiz 4 **General relativity** **Date 06-04-2026** **LC002**

Solution Key

1. The energy equation of a test particle in Schwarzschild space-time governed by mass M can be written as

$$\frac{\tilde{E}^2}{c^2} = \left(\frac{dr}{d\tau}\right)^2 + V^2$$

where

$$V^2(r) = c^2\left(1 - \frac{2GM}{c^2 r}\right)\left(1 + \frac{\ell^2}{r^2 c^2}\right)$$

and $\tilde{E} = E/m$ and ℓ is the angular momentum per unit test mass m.

(a) Using $V^2(r)$, identify the extra term that distinguishes from the effective potential of the central force problem. [1]

0.5 pt for correctly expanding $V^2(r)$. 0.5 pt for correctly writing V_{Newt} and obtaining the extra term: $-\frac{2GM\ell^2}{c^2 r^3}$.

(b) Find the radius of the innermost stable circular orbit (ISCO) in terms of R_s. [2]

Using $R_S = 2GM/c^2$

Evaluate $\frac{dV^2}{dr} = c^2\left(\frac{R_S}{r^2} - \frac{2\ell^2}{r^3 c^2} + \frac{3R_S\ell^2}{c^2 r^4}\right)$

0.5 pt for stating $\frac{dV^2}{dr} = 0$ for circular orbit and for obtaining roots of quadratic eqn.: $r = \frac{2\ell^2 \pm \sqrt{4\ell^4 - 12R_S^2\ell^2 c^2}}{2R_S c^2}$

0.5 pt for stable condition either from sketch or double derivative > 0

0.5 pt for obtaining the condition for innermost stable orbit as $\ell^2 = 3R_S^2 c^2$ (or $\ell^2 = 12M^2$ in geometric units).

0.5 pt for substituting ℓ in r and obtaining $r_{\text{ISCO}} = 3R_S$.

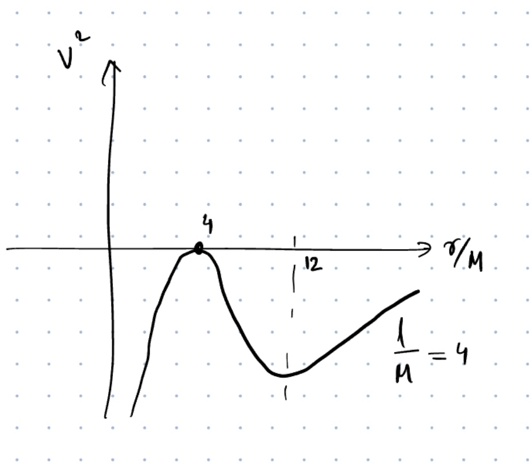
(c) For $\ell/M = 4$: [2]

i. Obtain the condition for close orbits.
Switch everything to geometric units.
0.5 pt for obtaining $r = 12M$ as the radius of stable circular orbit for $\ell/M = 4$.

0.5 pt for obtaining the condition for the bound orbits

$$V^2(12M) = \left(1 - \frac{2M}{12M}\right)\left(1 + \frac{(4M)^2}{(12M)^2}\right) = \frac{25}{27}$$

$$\sqrt{\frac{25}{27}} < \tilde{E} < 1 \tag{1}$$



ii. Under what condition, a test particle will directly fall into the BH.

0.5 pt for radius of unstable circular orbit $r = 4M$

0.5 pt for finding the condition.

$$\begin{aligned} V^2(4M) &= \left(1 - \frac{2M}{4M}\right)\left(1 + \frac{(4M)^2}{(4M)^2}\right) \\ &= \left(\frac{1}{2}\right)(1+1) = 1 \end{aligned}$$

$$\tilde{E} \geq 1$$